INCH-POUND

MS35690F
8 April 2005
Superseding
MS35690E
29 August 1984

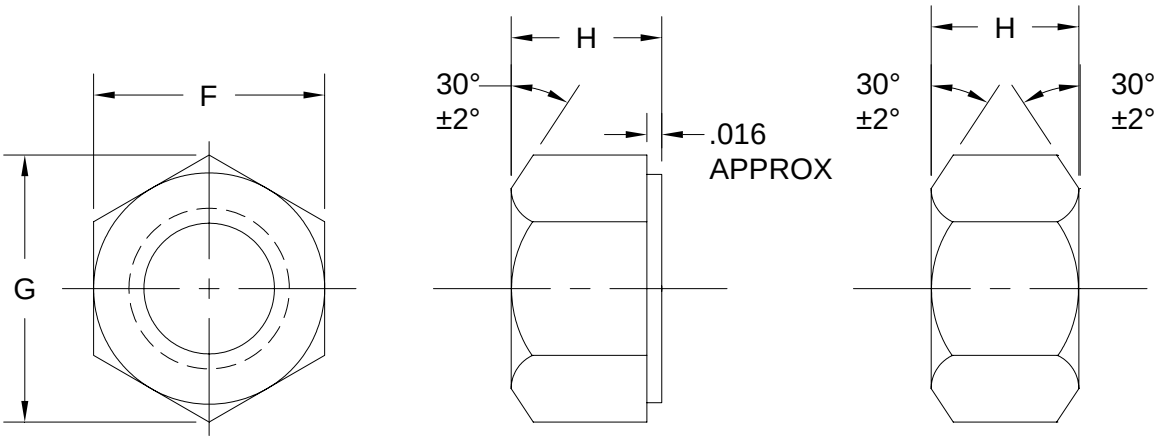
DETAIL SPECIFICATION SHEET

NUT, PLAIN, HEXAGON, UNC-2B AND UNF-2B

Reactivated after 8 April 2005 and may be used for new and existing designs and acquisitions.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and FF-N-836.



OPTIONAL DESIGNS

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TABLE I. DESIGN CONFIGURATIONS.

NOMINAL SIZE OR BASIC MAJOR DIAMETER OF THREAD	WIDTH ACROSS FLATS F		WIDTH ACROSS CORNERS G		THICKNESS H	
	MAX	MIN	MAX	MIN	MAX	MIN
1/4 - .2500	.4375	.428	.505	.488	.226	.212
5/16 - .3125	.5000	.489	.577	.557	.273	.258
3/8 - .3750	.5625	.551	.650	.628	.337	.320
7/16 - .4375	.6875	.675	.794	.768	.385	.365
1/2 - .5000	.7500	.736	.866	.840	.448	.427
9/16 - .5625	.8750	.861	1.010	.982	.496	.473
5/8 - .6250	.9375	.922	1.083	1.051	.559	.535
3/4 - .7500	1.1250	1.088	1.299	1.240	.665	.617
7/8 - .8750	1.3125	1.269	1.516	1.447	.776	.724
1 - 1.0000	1.5000	1.450	1.732	1.653	.887	.831
1 1/8 - 1.1250	1.6875	1.631	1.949	1.859	.999	.939
1 1/4 - 1.2500	1.8750	1.812	2.165	2.066	1.094	1.030
1 3/8 - 1.3750	2.0625	1.994	2.382	2.273	1.206	1.138
1 1/2 - 1.5000	2.2500	2.175	2.598	2.480	1.317	1.245
1 3/4 - 1.7500	2.6250	2.538	3.031	2.893	1.540	1.460
2 - 2.0000	3.0000	2.900	3.464	3.306	1.763	1.675
2 1/4 - 2.2500	3.3750	3.262	3.897	3.719	1.970	1.874
2 1/2 - 2.5000	3.7500	3.625	4.330	4.133	2.193	2.089
2 3/4 - 2.7500	4.1250	3.988	4.763	4.546	2.415	2.303
3 - 3.0000	4.5000	4.350	5.196	4.959	2.638	2.518

REQUIREMENTS:

1. Material: Steel, carbon, in accordance with procurement specification.
   Steel, corrosion resistant, types 303 (UNS S30300), 304 (UNS S30400), 305 (UNS S30500), 410 (UNS S41000), 416 (UNS S41600), 430 (UNS S43000) in accordance with ASTM A276 and ASTM A582.
   Brass, alloy 462 (UNS C46200) hard or 464 (UNS C46400) half-hard in accordance with ASTM B21.

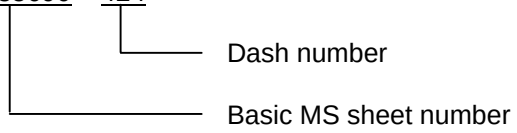
2. Protective coating and surface treatment:
   Cadmium plated, in accordance with procurement specification.
   Zinc coating, ASTM B633, Fe/Zn5 (5 μm thick), Type II.
   Phosphate coating, MIL-DTL-16232, Type Z, Class 2.
   Tin plated, .0001 thick in accordance with ASTM B545, Class A.
   Cleaned, descaled and passivated, in accordance with ASTM A380 (see Table IV).
   Nickel plated, in accordance with procurement specification (see Table II).

3. Threads: Threads shall be in accordance with FED-STD-H28/2. Acceptability of screw threads shall be in accordance with FED-STD-H28/20, System 21.

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4. Part number: The part number consists of the basic MS sheet number, followed by a dash number taken from Table II.

Example: MS35690 – 424



MS35690-424 indicates – Nut, Plain, Hexagon, Thread size .2500-28 UNF-2B; Steel, Carbon, Grade B, Uncoated.

5. Dimensions in inches.

6. Referenced documents shall be of the issue in effect on the date of invitation for bids or request for proposal, except that referenced adopted documents shall give the date of the issue adopted.

7. For design feature proposes, this standard takes precedence over procurement documents referenced herein.

8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

ACTIVE FOR NEW DESIGN
The following parts are reinstated for design after 27 September 1967.

Table II. Dash Numbers.

Thread Size	.2500-28UNF-2B	.3750-16-UNC-2B	.3750-24-UNF-2B
Material	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	424	604	624
Steel, Carbon, Grade C, Uncoated	427	607	627
Brass, Nickel Plated 1/	413	613	633

1/ New dash numbers added for nickel plated brass.

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INACTIVE PART NUMBERS FOR NEW DESIGN

The following parts are inactive for design after 30 September 1965.
Use for maintenance of existing equipment only

Table III.

Thread Size	.2500-20UNC-2B		.3125-18UNC-2B	.3125-24UNF-2B
Material	Dash No.		Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	404		504	524
Steel, Carbon, Grade C, Uncoated	407		507	527
Thread Size	.4375-14UNC-2B	.4375-20UNF-2B	.5000-13UNC-2B	.5000-20UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	704	724	804	824
Steel, Carbon, Grade C, Uncoated	707	727	807	827
Thread Size	.5625-12UNC-2B	.5625-18UNF-2B	.6250-11UNC-2B	.6250-18UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	904	924	1004	1024
Steel, Carbon, Grade C, Uncoated	907	927	1007	1027
Thread Size	.7500-10UNC-2B	.7500-16UNF-2B	.8750-9UNC-2B	.8750-14UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	1204	1224	1404	1424
Steel, Carbon, Grade C, Uncoated	1207	1227	1407	1427
Thread Size	1.0000-8UNC-2B	1.0000-12UNF-2B	1.1250-7UNC-2B	1.1250-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	1604	1624	1804	1824
Steel, Carbon, Grade C, Uncoated	1607	1627	1807	1827
Thread Size	1.2500-7UNC-2B	1.2500-12UNF-2B	1.3750-6UNC-2B	1.3750-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	2004	2024	2204	2224
Steel, Carbon, Grade C, Uncoated	2007	2027	2207	2227
Thread Size	1.5000-6UNC-2B	1.5000-12UNF-2B	1.7500-5UNC-2B	2.0000-4.5UNC-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	2404	2424	2804	3204
Steel, Carbon, Grade C, Uncoated	2704	2427	2807	3207
Thread Size	2.2500-4.5UNC-2B	2.5000-4UNC-2B	2.7500-4UNC-2B	3.0000-4UNC-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade B, Uncoated	3604	4004	4404	4804
Steel, Carbon, Grade C, Uncoated	3607	4007	4407	4807

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CANCELLED PART NUMBERS

The following parts are cancelled for design after 30 September 1965. For interchangeability see Tables V and VI of this standard and MS51967 thru MS51972.

Table IV.

Thread Size	.2500-20UNC-2B	.2500-28UNF-2B	.3125-18UNC-2B	.3125-24UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 401 402 403	 421 422 423	 501 502 503	 521 522 523
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 405 405	 425 426	 505 506	 525 526
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 408 409	 428 429	 508 509	 528 529
Steel, CRES Passivated	410	430	510	530
Brass Uncoated Tin plated	 411 412	 431 432	 511 512	 531 532
Thread Size	.3750-16UNC-2B	.3750-24UNF-2B	.4375-14UNC-2B	.4375-20UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 601 602 603	 621 622 623	 701 702 703	 721 722 723
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 605 606	 625 626	 705 706	 725 726
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 608 609	 628 629	 708 709	 728 729
Steel, CRES Passivated	610	630	710	730
Brass Uncoated Tin plated	 611 612	 631 632	 711 ---	 731 ---
Thread Size	.5000-13UNC-2B	.5000-20UNF-2B	.5625-12UNC-2B	.5625-18UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 801 802 803	 821 822 823	 901 902 903	 921 922 923
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 805 806	 825 826	 905 906	 925 926
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 808 809	 828 829	 908 909	 928 929
Steel, CRES Passivated	810	830	910	930
Brass Uncoated	811	831	911	931

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Table IV – Continued

Thread Size	.6250-11UNC-2B	.6250-18UNF-2B	.7500-10UNC-2B	.7500-16UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 1001 1002 1003	 1021 1022 1023	 1201 1202 1203	 1221 1222 1223
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 1005 1006	 1025 1026	 1205 1206	 1225 1226
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 1008 1009	 1028 1029	 1208 1209	 1228 1229
Steel, CRES Passivated	1010	1030	1210	1230
Brass Uncoated	1011	1031	1211	1231
Thread Size	.8750-9UNC-2B	.8750-14UNF-2B	1.0000-8UNC-2B	1.0000-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 1401 1402 1403	 1421 1422 1423	 1601 1602 1603	 1621 1622 1623
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 1405 1406	 1425 1426	 1605 1606	 1625 1626
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 1408 1409	 1428 1429	 1608 1609	 1628 1629
Steel, CRES Passivated	1410	1430	1610	1630
Brass Uncoated	1411	1431	1611	1631
Thread Size	1.1250-7UNC-2B	1.1250-12UNF-2B	1.2500-7UNC-2B	1.2500-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 1801 1802 1803	 1821 1822 1823	 2001 2002 2003	 2021 2022 2023
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 1805 1806	 1825 1826	 2005 2006	 2025 2026
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 1808 1809	 1828 1829	 2008 2009	 2028 2029
Steel, CRES Passivated	1810	1830	2010	2030
Brass Uncoated	1811	1831	2011	2031

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Table IV – Continued

Thread Size	1.3750-6UNC-2B	1.3750-12UNF-2B	1.5000-6UNC-2B	1.5000-12UNF-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 2201 2202 2203	 2221 2222 2223	 2401 2402 2403	 2421 2422 2423
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 2205 2206	 2225 2226	 2405 2406	 2425 2426
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 2208 2209	 2228 2229	 2408 2409	 2428 2429
Steel, CRES Passivated	2210	2230	2410	2430
Brass Uncoated	2211	2231	2411	2431
Thread Size	1.7500-5UNC-2B	2.0000-4.5UNC-2B	2.2500-4.5UNC-2B	2.5000-4UNC-2B
Material	Dash No.	Dash No.	Dash No.	Dash No.
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 2801 2802 2803	 3201 3202 3203	 3601 3602 3603	 4001 4002 4003
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 2805 2806	 3205 3206	 3605 3606	 4005 4006
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 2808 2809	 3208 3209	 3608 3609	 4008 4009
Steel, CRES Passivated	2810	3210	3610	4010
Brass Uncoated	2811	3211	3611	4011
Thread Size	2.7500-4UNC-2B	3.0000-4UNC-2B		
Material	Dash No.	Dash No.		
Steel, Carbon, Grade A, Uncoated CAD or Zinc optional Phosphate	 4401 4402 4403	 4801 4802 4803		
Steel, Carbon, Grade B, CAD or Zinc optional Phosphate	 4405 4406	 4805 4806		
Steel, Carbon, Grade C, CAD or Zinc optional Phosphate	 4408 4409	 4808 4809		
Steel, CRES Passivated	4410	4810		
Brass Uncoated	4411	4811		

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INACTIVE PART NUMBERS FOR NEW DESIGN

After 30 September 1965, the following canceled nuts are superseded by the inactivated nuts indicated below. However, the canceled nuts should be used until existing stocks are depleted. Then use the superseding inactivated nuts for maintenance purposes only. For new design and engineering use MS51967 thru MS51972.

Table V. Interchangeability table of canceled and superseding part numbers.

Part Numbers		Part Numbers		Part Numbers	
Canceled	Superseded by	Canceled	Superseded by	Canceled	Superseded by
MS35690-401	MS35690-404	MS35690-1001	MS35690-1004	MS35690-2021	MS35690-2024
-501	-504	-1021	-1024	-2201	-2204
-521	-524	-1201	-1204	-2221	-2224
-701	-704	-1221	-1224	-2401	-2404
-721	-724	-1401	-1404	-2421	-2424
-801	-804	-1421	-1424	-2801	-2804
-821	-824	-1601	-1604	-3201	-3204
-901	-904	-1621	-1624	-3601	-3604
-921	-924	-1801	-1804	-4001	-4004
		-1821	-1824	-4401	-4404
		-2001	-2004	-4801	-4804

The following canceled nuts are superseded by the reinstated nuts indicated below.

Table VI. Interchangeability table of canceled and superseding part numbers.

Part Numbers		Part Numbers		Part Numbers	
Canceled	Superseded by	Canceled	Superseded by	Canceled	Superseded by
MS35690-421	MS35690-424 -427	MS35690-601	MS35690-604 -607	MS35690-621	MS35690-624 -627

MILITARY INTEREST

Custodians:
Navy - SH
Air Force - 99

Preparing activity:
DLA - IS

(Project 5310-2721)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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